

Question Number	Answer	Additional Guidance	Mark
3a	there is no (significant) difference between the mitotic index at different oxygen concentrations (after 2 days) (1)	Candidates will express this in different ways	1

Question Number	Answer	Additional Guidance	Mark
3b	An answer that includes the following: - suitable table format with correct column headings and units (1) - all data correctly entered (1) - means correctly calculated (1)	Allow mitotic index % or au not if units in table 5.7 and 4.0 Do not allow 4	3

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3c	An answer that includes the following: - bar graph with linear scale starting at zero and axes labelled, with units (1) - means plotted correctly (1) - range bars plotted correctly (1)	mean mitotic index allow % as unit for mitotic index oxygen % ECF using the means from 3b Allow tolerance of half a small square 3.6 to 4.4 5.1 to 6.2	3

Question Number	Answer	Additional Guidance	Mark
3di	- correct substitution of given $(SA)^2$ and $(SB)^2$ (1) - correct answer (1)	Allow ECF if incorrect means used (denominator is 0.1499923) 11.33 / 11.3 Ignore minus sign	2

Question Number	Answer	Additional Guidance	Mark
3dii	An answer that includes the following points: (pea seeds grown) under 20% oxygen have a larger mitotic index than those grown under 5% oxygen (1) - the calculated value of t (11.33) is more than the critical value 2.12 (1) (therefore) reject the null hypothesis, the difference in the mitotic index at 5% oxygen compared to 20% oxygen is significant (1)	Allow any quoted t value and a following correct statement for MP2 and MP3 Allow 2.92 at 1% significance Allow different concentrations of oxygen	3

Question Number	Answer	Additional Guidance	Mark
3e	An answer that includes two of the following points: • use different temperatures (1) • use different oxygen concentrations (1) • use different times (1) • use different plant species (1)	Ignore repeat the expt Allow within the 5-20% range or a wider range Allow variety / cultivar / type	2

(Total for question 3 = 14 marks)

